Postura – Step 6: CPE Review Screen

Developer instructions - March 2026

1 PURPOSE

CPE = Certified Professional Ergonomist. This is the trained ergonomist who reviews and signs off each assessment before it is approved. The employee takes the photo and fills in the questions. The CPE then reviews all collected data on this screen and gives a formal decision. This screen must never require the CPE to re-enter any data. Everything is pre-filled and read-only.

2 POSITION IN APP FLOW

Step	Screen	Who
1	Consent	Employee
2	Pain duration / pattern	Employee
3	Work Pattern (hours, device, mouse)	Employee
4	Your Workstation (7 questions)	Employee
5	Photo capture -> ROSA computed	Employee
6	Results screen	Employee
7 >	CPE Assessment Review ← THIS SCREEN	CPE Ergonomist

3 WHAT ALREADY EXISTS — DO NOT CHANGE

The following elements are already built on the CPE review screen. Keep all of them exactly as they are.

- Employee name + ID (header)
- Screen title: CPE Assessment Review
- Subtitle: Review data and complete checklist to approve
- Desk Info block: Desk ID / Location + Role
- Pain & Symptoms block: body part + duration + pain intensity score
- 2 photo thumbnails (will be updated — see Section 5)
- 4 approval checkboxes (keep exactly as is)
- Remote / Live review toggle
- Decision dropdown (Approved / Flag / Reject)
- Comment text field
- Add Signature / Upload Signature
- Submit Review button (disabled until all 4 checkboxes ticked + decision set)

4 WHAT CHANGES — TWO BLOCKS REPLACED / ADDED

4.1 Replace ISO circle block with ROSA Score block

Remove: the circular ISO compliance percentage (17% circle with ring).

Replace with: a ROSA Score block as described below.

4.2 Add Workstation Inputs block

Insert a new read-only block directly below the ROSA Score block and above the Pain & Symptoms block. This block shows all 10 inputs collected before the photo: 3 from the Work Pattern screen (hours, device, mouse) and 7 from the Workstation Questions screen. See Section 6 for the full list.

5 ROSA SCORE BLOCK — FULL SPECIFICATION

Block title: ROSA Assessment

Show the final ROSA score prominently with color-coded risk tier. Below it, show the 5 sub-scores so the CPE can verify each component.

5.1 Final score display

Final ROSA Score	7 / 10 RED - Immediate action required

Color logic — same as existing risk tiers in the app:

ROSA Score	Color	Label
1 – 3	GREEN	Low risk — monitor only
4 – 5	ORANGE	Medium risk — reassess in 3 months
6 – 10	RED	High risk — immediate action required

5.2 Sub-scores (read-only rows)

Chair score	4	From knee angle + trunk + armrest
Monitor score	3	From neck landmarks + dual screen
Keyboard score	2	From wrist vs elbow position
Mouse score	2	From arm reach + mouse type question
Peripherals score	5	From Table D: monitor + keyboard + mouse

Sub-scores are read-only. The CPE cannot edit them. If a sub-score looks wrong, the CPE uses the comment field to flag it before approving.

6 WORKSTATION INPUTS BLOCK — FULL SPECIFICATION

Block title: Workstation Inputs

7 read-only fields showing exactly what the employee answered before the photo. The CPE uses this to verify checkbox 2 (workstation equipment correctly identified) and checkbox 3 (ROSA score calculation is

verified).

Hours at desk per day	4+ hours
Device setup	Dual Screen
Mouse type	Standard mouse
Q1 — Chair height adjustable	No
Q2 — Leg room under desk	Yes
Q3 — Lumbar support	No
Q4 — Monitor distance	40–75 cm
Q5 — Feet flat on floor	No
Q6 — Monitor directly in front	Yes
Q7 — Chair has armrests	Yes

All fields are display-only. The CPE cannot edit any of these values. If an answer looks inconsistent with the photo (e.g. employee said Yes to lumbar support but the photo shows a stool), the CPE flags this in the comment field.

7 PHOTOS BLOCK — UPDATED SPEC

Currently: 2 raw photo thumbnails side by side.

Change to: left photo = raw, right photo = same image with MediaPipe landmarks drawn on top.

Build the full screen layout and design now — all blocks, read-only fields, checkboxes and layout. For the annotated photo drawing only: confirm coordinate format with Mawais after Step 2 delivery. Ask him: does the landmark object deliver normalized (0-1) or pixel coordinates? Everything else on this screen can be built immediately with hardcoded test values.

7.1 Annotated photo — what to draw

Landmark	Index	Lines to draw
Left ear	7	Ear -> shoulder (neck line)
Left shoulder	11	Shoulder -> elbow, shoulder -> hip
Left elbow	13	Elbow -> wrist (arm line)
Left wrist	15	End of arm line
Left hip	23	Hip -> knee (trunk/leg)
Left knee	25	Knee -> ankle
Left ankle	27	End of leg line

7.2 How to generate the annotated photo

• During photo capture, MediaPipe already outputs x,y coordinates for all 7 landmarks.

• After capture, draw dots (radius 5px) at each landmark position on the photo.

• Draw lines connecting: ear->shoulder, shoulder->hip, hip->knee, knee->ankle, shoulder->elbow, elbow->wrist.

• Color: use the app teal (#01696F) for all dots and lines.

• Line thickness: 2px.

• Save the annotated version alongside the raw photo. Both stored. Both sent to CPE screen.

• The CPE uses the annotated photo to verify checkbox 1: Posture landmarks are valid and accurate.

8 APPROVAL CHECKBOXES — KEEP AS IS

The 4 checkboxes remain exactly as currently built. No label changes. No order changes. Submit Review stays disabled until all 4 ticked + decision set.

[x] Posture landmarks are valid and accurate

[ ] Workstation equipment is correctly identified

[ ] ROSA score calculation is verified

[ ] Recommendations are professionally appropriate

Checkbox 1 -> CPE verifies using the annotated photo (Section 7). Checkbox 2 -> CPE verifies using the Workstation Inputs block (Section 6). Checkbox 3 -> CPE verifies using the ROSA sub-scores (Section 5). Checkbox 4 -> CPE verifies using the exercise/equipment recommendations on the results screen.

9 BACKEND DATA NOTE — IMPORTANT FOR FUTURE MATRIX

Store ROSA score and VAS pain scores together under the same assessment_id with a shared timestamp. Do NOT store them in separate tables without a shared key. This is required for the future combined ROSA + VAS risk matrix (Red/Orange/Green/Yellow tiers). If stored separately, joining them later requires a costly migration.

10 FINAL SCREEN LAYOUT — TOP TO BOTTOM

#	Block	Change?
1	Header: Employee name + ID + timestamp	No change
2	ROSA Assessment (score + sub-scores)	NEW — replaces ISO circle
3	Workstation Inputs (7 read-only fields)	NEW — insert here
4	Desk Info	No change
5	Pain & Symptoms	No change
6	Photos: raw + annotated side by side	Updated — add annotated photo
7	Approval checkboxes (4)	No change
8	Remote / Live toggle	No change

9	Decision dropdown	No change
10	Comment field	No change
11	Add Signature	No change
12	Submit Review button	No change

11 OCCLUSION FALLBACK — FOR INFORMATION ONLY

This section is FOR INFORMATION ONLY. The occlusion fallback logic is built by Mawais (Upwork developer) as part of Steps 1-3, not by you. You do not need to implement any code from this section. It is included here so the CPE review screen developer understands why the annotated photo may occasionally show a fallback ear-to-shoulder line instead of a direct ear landmark dot.

11.1 Neck landmark occlusion fallback

Condition: long hair covers the neck area, blocking clean neck landmark detection.
Trigger: MediaPipe visibility score for ear landmark (index 7) < 0.50.

Item	Detail
Normal path	Use ear.y - nose.y for neck flexion calculation as specified in score()
Fallback path	If ear visibility < 0.50: use direct ear-to-shoulder vector to estimate neck angle. Draw a line from ear landmark to shoulder landmark and calculate the angle from vertical. This is the same geometry already used in the annotated photo drawing.
Why this works	Ear and shoulder are proximal landmarks — significantly more reliable under occlusion than distal joints. Published reliability study (PubMed 2022) confirms shoulder/elbow angles from pose estimation have ICC > 0.60. The ear-to-shoulder line approximates the cervical spine vector, which is the clinical basis for neck flexion in ROSA.
Source	ArXiv 2025: proximal joints degrade less under occlusion. PubMed 2022: shoulder landmark ICC > 0.60 in pose estimation. MediaPipe Tasks API: visibility score 0-1 per landmark, 0.50 is Google default threshold.

11.2 Dart implementation

```
// NECK OCCLUSION FALLBACK // If ear landmark visibility < 0.50, use ear-to-shoulder vector instead if (ear.visibility < 0.50) { // Fallback: angle of ear->shoulder line from vertical double dx = (shoulder.x - ear.x).abs(); double dy = (shoulder.y - ear.y).abs(); double fallbackAngle = atan2(dx, dy) * 180 / pi; // Map to ROSA neck score using same thresholds if (fallbackAngle > 20) monitorScore = 2; else if (fallbackAngle > 35) monitorScore = 3; else monitorScore = 1; } else { // Normal path - use existing score() logic }
```

11.3 Future occlusion testing

After Step 8 (CPE blind study) is complete, run a secondary stress test with photos containing: long hair (neck occlusion), baggy clothing (elbow/hip occlusion), low contrast lighting. Target: congruence +/-1 does not drop below 70% on occluded cases. This is a future step — do not block Steps 4-7 on this.

12 DEV TIME ESTIMATE

Task	Time

Remove ISO circle block	15 min
Build ROSA score block (final score + 5 sub-scores + color logic)	1.5 hours
Build Workstation Inputs block (10 read-only rows)	1 hour
Display annotated photo alongside raw photo (photo already generated by Mawais pipeline)	30 min
Total Step 6	~2-3 hours
Note: annotated photo generation, backend storage and occlusion fallback are Mawais tasks — not included here	

Sources

1. Cornell ROSA 2011 Instructions - ergo.human.cornell.edu/CUErgoTools/ROSA/ROSA-Instructions-2011-2012.pdf
2. Moriguchi et al. 2022, BMC Musculoskeletal Disorders - ROSA validation study - pmc.ncbi.nlm.nih.gov/articles/PMC9160176/
3. Combined ROSA + VAS risk matrix rationale - PLoS ONE 2019 - pmc.ncbi.nlm.nih.gov/articles/PMC6890250/
4. Occlusion in pose estimation - ArXiv 2025 benchmark - arxiv.org/html/2504.10350v2
5. Pose estimation reliability for joint angles - PubMed 2022 - pubmed.ncbi.nlm.nih.gov/36131313/